

9(a)	<i>any seven from:</i> 1 double stranded DNA ; 2 have, restriction site / target sequence for restriction enzyme ; 3 allows <u>gene</u> (for cloning) to be inserted ; 4 small ; 5 plasmid can, enter / transform, (host) cell / bacterium ; 6 circular ; 7 stable ; 8 contain, (named) marker <u>genes</u> / (named) <u>genes</u> for antibiotic resistance ; 9 used to identify, GM / transformed / recombinant, (bacterial) cells ; 10 replicate, fast / independently (of host cell replication) ; 11 get many copies of cloned gene ; 12 AVP ;	7
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Question	Answer	Marks
9(b)	any eight from: <i>restriction endonucleases</i> 1 cut <u>DNA</u> ; 2 at, restriction site / specific (base) sequence / target sequence ; 3 <i>ref. to</i> palindrome ; 4 give, sticky / blunt, ends ; <i>ligases</i> 5 join gene to, plasmid / (named) vector ; 6 seal (sugar-phosphate) backbone ; 7 make, phospho(di)ester / covalent, bonds ; 8 make recombinant, DNA / plasmid / vector ; <i>reverse transcriptase</i> 9 makes cDNA ; 10 from mRNA ; 11 <i>ref. to</i> DNA single-stranded initially ; 12 AVP ;	8